

AIMScribe v3 — Running It In The Cloud, Permanently

Point-by-point justification: the disk, the databases, the dashboard, and what it costs for ever · AIMS LAB · 4
October 2026

0 · A correction to the previous version

The last plan said the server would need "under 60 GB, for ever" once the audio moved to object storage. **That was wrong, and you were right to challenge it.** It counted the operating system, the databases and one consultation in flight, and left out six things that are not optional:

the buffer needed when object storage is unreachable · PostgreSQL's write-ahead log · local backup staging · Docker images and build cache · log growth · and the fact that the databases **grow for ever** while everything else is steady state.

The corrected figure is **200 GB minimum, 250 GB recommended, and it has to be able to grow.** Section 3 shows the line-by-line working. **100 GB is not enough even after the change.**

1 · Where every piece of data lives

You asked specifically where the databases, the patient information, the audio tracking and the dashboard live. All four are on the **server**, not in object storage. Only the audio leaves.

What	Where it lives	How big	Grows?
aims_recordings the audio-tracking database	PostgreSQL, on the server's disk. Every recording, every piece, the integrity chain, the purge receipts, the audit log. No patient names, by design	~30 GB per year	for ever
aims_clinical patient information	PostgreSQL, on the same server, different database and different login. Demographics, paramedic readings, prescriptions, diagnoses, every CMED message kept verbatim	~10 GB per year	for ever
The dashboard	Served by the API container on the server. It is one HTML file; it reads aims_recordings only and holds no credential for the clinical database, so it cannot show a patient's name	a few MB	no
Working files	The server's disk: pieces downloaded, joined, compressed, encrypted — one consultation at a time, then deleted	~3 GB	no
The WAV audio	Object storage (hot). Read in seconds	2–3 TB rolling	capped
The FLAC copies	Glacier Deep Archive (cold). The permanent record	6.2 TB per year	for ever

The databases are the part that must never be lost and must always be fast. They are small compared to audio but they are the index to everything: without them the audio files are a folder of names nobody can interpret. They stay on the server's disk, on SSD, and are backed up nightly to object storage.

2 · The numbers for permanent running

Figure	Value
Consultations per clinic day	277
Clinic days per year	312 (six days a week)
Consultations per year	86,400
Average file	120 MB — which is 24 minutes at this format
Audio per day	33 GB
WAV per year	10.4 TB
FLAC per year (60% of WAV)	6.2 TB
Database growth per year	~40 GB for both together

Here is the thing your 2–3 TB figure runs into. At 33 GB a day, 2 TB of raw WAV is **60 clinic days** and 3 TB is **90**. That is two to three and a half months of recording, not a permanent store. Running for ever produces 10.4 TB of WAV every year.

So "2–3 TB at most for raw WAV" can only mean one thing, and happily it is the right thing: **a rolling window**. Keep the most recent two to three months of WAV hot and readable; everything older exists as FLAC in the deep archive.

This costs you nothing in quality, and that is the key fact. FLAC is lossless. It decodes back to exactly the same samples — the system already proves this on every file, by decoding the FLAC again and comparing it to the archived WAV before the copy is accepted. A WAV deleted from the hot window can be rebuilt byte for byte from its FLAC. You are not throwing anything away; you are choosing how fast it is to reach.

3 · The disk budget, line by line

This is the working you asked for. Every line is a thing that must fit on the server's own disk at the same time.

#	What	Size	Why it cannot be left out
1	Operating system	20 GB	Ubuntu, packages, kernels
2	Docker images and build cache	15 GB	Eight containers: Postgres, Redis, the API, the worker (with the audio codec), the gateway, PgBouncer, the viewer, CMED's page. Build cache grows unless pruned
3	Both databases, first year	40 GB	Measured at roughly 0.3–0.5 MB per recording across both. This grows by about 40 GB every year and never shrinks
4	PostgreSQL write-ahead log	10 GB	Every change is written here before the tables. Needed for crash safety and for point-in-time recovery
5	Backup staging	10 GB	The nightly dump is written locally, compressed, encrypted, then uploaded. Keep two nights on disk
6	Working space	3 GB	Pieces downloaded, joined, compressed to FLAC, encrypted — several consultations at once
7	Offload buffer	50 GB	The one most easily forgotten. If object storage is unreachable, finished recordings pile up here at 33 GB a day. 50 GB buys about a day and a half before the system pauses
8	Logs	5 GB	Eight containers, rotated, plus the database's own logs
9	Enforced free space	20 GB	The archive worker refuses to write below this, on purpose, so a full disk never corrupts a recording
10	Filesystem reserve and headroom	~30 GB	ext4 reserves 5%, and no disk should run above 80% full
Year 1 total		203 GB	so: 250 GB, with room to grow

Year 3	~283 GB	the databases have added 80 GB
Year 5	~363 GB	400 GB, or move the databases to managed storage

Line 7 is the one that decides whether the system runs without interruption. Everything else is steady. That buffer is what lets a storage outage, a credential expiry or a network problem pass without the clinics noticing. Make it bigger if you can — disk is the cheapest part of this.

4 · The revised server

	Minimum	Recommended	Why
Cores	4	8	Compressing to FLAC is the real work: about 11 hours of one core per day, spread over 24
Memory	8 GB	16 GB	Everything together uses about 1 GB in normal running, with ceilings at 3.4 GB. 16 GB leaves room for the database to cache and for the working area to live in memory rather than on disk
Disk	200 GB SSD	250 GB SSD, expandable	Section 3. It must be able to grow, because the databases do
Network	100 Mbit/s	1 Gbit/s	33 GB in and about 53 GB out per day, spread across the day

So the AIMS LAB machine is close, but its disk is not. 8 cores and 8 GB are fine. 100 GB is half what is needed in year one and a third of what is needed by year five. If that machine can take a second disk, or a volume can be attached to it, it becomes the cheapest option by far. If it cannot, rent.

5 · What it costs to run for ever

Two things grow and one does not. The server is flat. The hot window is capped by you. **Only the deep archive grows**, and it is the cheapest thing here.

Option A — keep the AIMS LAB server, add a disk

Item	Detail	Year 1	Year 3	Year 5
Server	already owned	\$0	\$0	\$0
Extra disk	250–400 GB, one-off	~\$40	—	~\$40
Hot WAV	Cloudflare R2, 3 TB rolling	\$45/mo	\$45/mo	\$45/mo
Cold FLAC	Glacier Deep Archive, grows 6.2 TB/yr	\$6/mo	\$19/mo	\$31/mo
Monthly, end of year		~\$51	~\$64	~\$76
That year's total		~\$400	~\$700	~\$880

Option B — DigitalOcean, everything rented

Item	Detail	Year 1	Year 3	Year 5
Server	4 vCPU / 16 GB / 200 GB	\$84/mo	\$84/mo	\$84/mo
Extra volume	100–200 GB as databases grow	\$10/mo	\$15/mo	\$20/mo
Hot WAV	Cloudflare R2, 3 TB rolling	\$45/mo	\$45/mo	\$45/mo
Cold FLAC	Glacier Deep Archive	\$6/mo	\$19/mo	\$31/mo
Backups	server snapshots	\$17/mo	\$17/mo	\$17/mo
Monthly, end of year		~\$162	~\$180	~\$197
That year's total		~\$1,800	~\$2,100	~\$2,300

Prices are approximate, October 2026, and should be confirmed with each provider. **Over five years Option A costs roughly \$3,000 and Option B roughly \$10,000** — the difference is almost entirely the rented server.

6 · Point-by-point justification

#	Decision	Why	What happens if you do the opposite
1	Audio goes to object storage, not a disk	Twenty times cheaper per terabyte, never fills, no RAID to buy, and it is replicated by the provider	A 10 TB disk a year, each one a single copy you must mirror yourself
2	The databases stay on the server's SSD	They are small, they must be fast, and every query the dashboard makes hits them. Object storage cannot serve a database	The dashboard becomes unusable and the archive worker slows to a crawl
3	The hot WAV window is capped and rolling	It bounds a cost that would otherwise grow by 10.4 TB a year, and FLAC means nothing is lost by ageing files out	\$150 a month by year three, rising for ever, for files nobody is reading
4	FLAC goes to the deep archive and stays for ever	It is the permanent record, it is lossless, and at \$1 per TB per month it is the cheapest durable storage that exists	You keep expensive hot copies of data you touch once a year
5	Deep archive is never the working store	Getting a file back takes 12–48 hours and costs extra each time	A researcher waits two days for one consultation
6	The disk has a 50 GB offload buffer	A storage outage then passes unnoticed instead of stopping the clinics	Any cloud hiccup becomes a clinical interruption within hours
7	One server, not two	What keeps the clinics recording through an outage is the buffer on each laptop, not a second server. Two servers double cost and more than double complexity	You pay twice for resilience you already have, and have two things to patch
8	Nightly database backups, off the machine	The databases are the index to everything. Losing them makes the audio uninterpretable	A disk failure costs you the ability to find anything, even though the audio survives
9	Transit and hot archive at the same provider	Simpler, and downloads between them are free	Paying egress twice, and two sets of credentials to rotate
10	Restore tested on day one, then monthly	A copy nobody has read is not a backup	You discover the copy was unreadable on the day you need it

7 · What runs where, in one picture

Layer	Component	Lives on	If it stops
Clinic	AIMScribe agent, one per room	the doctor's laptop	That room stops recording. Others continue
Clinic	CMED's page	the doctor's laptop, in a browser	No consultation can be opened
Transit	Audio pieces	object storage	Laptops hold up to 13 h, then stop
Server	The API — grants, Channel B, the dashboard	the server	Recording continues on the laptops; nothing is archived
Server	aims_recordings — audio tracking	the server's SSD	Nothing can be confirmed or archived
Server	aims_clinical — patient data	the server's SSD	Recordings continue but go unconfirmed
Server	Redis — the job queue	the server's SSD	Transcription queues stall
Server	Archive worker — joins, compresses, uploads	the server	Recordings queue in the offload buffer
Storage	Hot WAV, rolling window	object storage	Researchers cannot read audio; recording is unaffected
Storage	Cold FLAC, permanent	deep archive	Nothing immediate. It is insurance

8 · What has to be built before any of this

	Work	Why it is needed	Size
1	Offload the WAV to object storage after the copy is verified, then delete it locally	Without it the disk fills in three days and nothing else in this plan matters	a few days
2	Age the hot window — delete WAVs older than the chosen window, but only where a verified FLAC exists	This is what keeps the hot cost flat for ever	a day
3	Point the dashboard and the restore tool at the new location	So a recording can still be found and fetched	a day
4	Alerting on the offload buffer, failed jobs and silent recorders	The buffer is only useful if somebody is told before it fills	a day

Item 2 deserves care. It deletes patient audio, so it must refuse unless the FLAC exists, has been verified, and the database says so. Written carelessly it is the one piece of this system that could lose a recording. Written properly it is what makes permanent operation affordable.

9 · One page to remember

Question	Answer
Can 60 GB run this?	No. That figure was wrong. 200 GB minimum, 250 GB recommended, and it must be able to grow
Where are the databases?	On the server's SSD. Both of them — audio tracking and patient data, separate databases, separate logins
Where does the dashboard run?	On the server, served by the API. It reads the audio-tracking database only and cannot see a patient's name
How fast do the databases grow?	About 40 GB a year, for ever

Is 2–3 TB of raw WAV enough?	Only as a rolling window of 2–3 months. Permanent running makes 10.4 TB a year
Does ageing out WAVs lose anything?	No. FLAC is lossless and verified by decoding; the WAV can be rebuilt exactly
Cheapest permanent setup	Keep the AIMS LAB server, add a 250–400 GB disk, R2 for the hot window, Glacier for the cold archive
What does that cost?	About \$51 a month in year one, \$76 by year five. Roughly \$3,000 over five years
What if we rent everything?	About \$162 a month rising to \$197. Roughly \$10,000 over five years
What grows for ever?	Only the deep archive and the databases — and both are cheap. The hot window is capped by you
Biggest thing still to build	The offload step, then the ageing-out step
Biggest risk	Not storage. It is nobody noticing when something stops

Prepared by AIMS LAB. Prices are indicative, October 2026, and should be confirmed with each provider before purchase.